Cardinality-Constrained KV Cache Quantization: Preventing 2-Bit Compounding Failures on Hardware-Aligned Bit-Widths

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Abstract

Per-head mixed-precision KV cache quantization promises superior quality–compression trade-offs over uniform bit-width allocation. However, on hardware-aligned non-consecutive bit-widths $\{2,4,8,16\}$, we discover a *universal failure mode*: 2-bit quantization distortions compound non-additively through the residual stream, causing *all* tested unconstrained solvers—greedy and MCKP—to over-allocate 2-bit heads and underperform naive uniform 4-bit across 8/8 diverse architectures (degradation 1.07×–3.31× baseline). A simple cardinality constraint ($\leq k$ heads at 2-bit) universally prevents this over-allocation and, surprisingly, makes solver choice irrelevant: constrained greedy and constrained MCKP produce *identical* bit allocations (Δ=0.0000 PPL). Under moderate compression (4×), only 1/8 architectures benefits from non-uniform allocation; the remaining 7/8 should use uniform 4-bit. A two-metric diagnostic (concentration $\gtrsim$0.50 AND $t_1/t_5 \lesssim$0.70) correctly classifies all 9 tested models. At 8K context length, uniform 4-bit degrades 11.15× baseline while constrained MCKP degrades only 1.23×—a 9× gap in degradation rate—demonstrating critical deployment value.

1 Introduction

KV cache memory is the primary bottleneck for long-context LLM inference. Per-head mixed-precision quantization allocates different bit-widths to different attention heads based on their sensitivity, promising better quality–compression trade-offs than uniform allocation. Prior work such as RateQuant [1] provides rate-distortion optimal allocation on consecutive integer bit-widths $\{2,\ldots,6\}$, leveraging polymatroid structure for provably optimal greedy solutions.

However, practical deployment requires *hardware-aligned* bit-widths $\{2,4,8,16\}$ supported by efficient GPU kernels. On these non-consecutive bit-widths, we discover that:

1. **2-bit compounding is universal.** Independent per-head profiling underestimates joint distortion when multiple heads are simultaneously quantized to 2-bit. Per-head profiling predicts joint PPL≈11 but actual joint 2-bit PPL is 32–37 (3× prediction error). All unconstrained solvers consequently over-allocate 2-bit heads and underperform uniform 4-bit across 8/8 tested architectures.

2. **A cardinality constraint is the universal fix—and makes solver choice irrelevant.** Capping the number of 2-bit heads at $k \leq 8$ prevents compounding across all architectures. Under this constraint, greedy and MCKP produce identical allocations (Δ=0.0000 PPL), eliminating solver complexity.

3. **Non-uniform benefit is rare.** Under moderate (4×) compression, only 1/8 architectures—those with a "spread" sensitivity pattern—benefits from non-uniform allocation. A two-metric diagnostic correctly predicts this for all 9 tested models, providing practitioners a simple check before investing in profiling.

Our contributions are: (i) the first systematic characterization of 2-bit compounding effects in per-head KV cache quantization, (ii) a universal constraint fix that eliminates solver complexity, and (iii) the first empirical study of *when* non-uniform allocation is beneficial, correcting the implicit assumption that it always helps.

2 Related Work

KV Cache Quantization. KIVI [2] applies per-channel group quantization with residual storage, achieving near-lossless 4-bit compression. Our work is orthogonal: KIVI optimizes *within-bit* quantization quality via group granularity, while we optimize *cross-bit* per-head allocation and characterize its failure modes. The two approaches are complementary and can be combined.

Rate-Distortion Allocation. RateQuant [1] formulates per-head allocation as rate-distortion optimization, proving greedy optimality on consecutive integers $\{2,\ldots,6\}$ via polymatroid structure (Proposition 6). On hardware-aligned $\{2,4,8,16\}$, this structure breaks: marginal gain per bit between $2\to4$ and $4\to8$ is non-monotone, and greedy over-allocates 2-bit heads (68/112 = 60.7% on Qwen, PPL 32.45 vs optimal $\sim$10.91). RateQuant acknowledges compounding effects (Appendix M) but does not address them. Our cardinality constraint fixes RateQuant's greedy, achieving identical results to MCKP.

Mixed-Precision Quantization. Weight quantization literature extensively studies per-layer or per-group mixed-precision allocation. KV cache quantization differs: attention heads exhibit architecture-specific sensitivity patterns that determine whether non-uniform allocation helps at all—a question not previously studied.

3 Method

3.1 Problem Formulation

Given a model with H KV heads and bit-width options $\mathcal{B} = \{2, 4, 8, 16\}$, we seek an assignment $b_h \in \mathcal{B}$ for each head h minimizing total distortion subject to a memory budget:

$$\min_{\{b_h\}} \sum_{h=1}^{H} d_h(b_h) \quad \text{s.t.} \quad \sum_{h=1}^{H} b_h \leq B \tag{1}$$

where $d_h(b_h)$ is the per-head distortion measured by the change in perplexity when head h is quantized to b_h bits (all other heads at full precision).

This is a standard Multiple-Choice Knapsack Problem (MCKP). Independent per-head profiling assumes distortions are additive: $D(\{b_h\}) = \sum_h d_h(b_h)$. We show this assumption fails when multiple heads are assigned 2-bit simultaneously.

3.2 The 2-Bit Compounding Effect

When k heads are simultaneously quantized to 2-bit, the joint distortion exceeds the sum of individual distortions:

$$D_{\text{joint}}(k) \gg \sum_{h \in S_k} d_h(2) \tag{2}$$

We characterize this non-additivity empirically via a compounding sweep: varying k from 0 to H while measuring actual PPL. On Qwen2.5-7B, profiling predicts joint PPL$\approx$11 at k=112 but actual PPL is 98.75 (9$\times$ prediction error).

3.3 Cardinality-Constrained Allocation

We augment the MCKP with a cardinality constraint:

$$|\{h : b_h = 2\}| \leq k \tag{3}$$

This constraint limits the number of 2-bit heads, preventing compounding. The constrained problem remains MCKP-solvable. We determine k via a small-scale compounding test: sweep k on a calibration set and select the largest k with $< 2.5\%$ degradation. Empirically, $k \leq 8$ is universally safe across all tested architectures.

3.4 Sensitivity Distribution Diagnostic

To determine whether non-uniform allocation is beneficial *a priori*, we propose a two-metric diagnostic based on the per-head sensitivity profile $\{d_h(2)\}$:

- **Top-5% concentration**: fraction of total sensitivity in the top 5% of heads. Threshold: $\gtrsim 0.50$.
- **Dominance ratio** t_1/t_5: ratio of the single most sensitive head to the sum of the top-5 heads. Threshold: $\lesssim 0.70$.

The joint condition (concentration $\gtrsim 0.50$ AND $t_1/t_5 \lesssim 0.70$) identifies "spread" patterns where multiple heads have high but differentiated sensitivity, enabling MCKP to exploit cross-bitwidth gradients. A "spike" pattern (high concentration but high t_1/t_5) indicates a single dominant outlier that cannot be exploited. A "flat" pattern (low concentration) indicates homogeneous sensitivity where uniform allocation is optimal.

4 Experiments

4.1 Setup

We evaluate on 8 architectures spanning GQA ratios from 1:1 to 8:1: Qwen2.5-7B (7:1, 112 KV heads), Yi-1.5-9B (8:1, 192 heads), Llama-3.1-8B (4:1, 256 heads), Mistral-7B-v0.3 (4:1, 256 heads), Gemma-2-9B (2:1, 336 heads), Pythia-1.4B (MHA, 384 heads), Qwen2.5-14B (7:1, 384 heads), and Phi-3-medium (400 heads). Profiling and evaluation use WikiText-2 at 4× compression (budget=0.25) unless stated otherwise. High-resolution profiling uses 64 sequences × 2048 tokens.

4.2 Unconstrained Solvers Universally Fail

Table 1 shows that unconstrained allocation (both greedy and MCKP) underperforms uniform 4-bit across all tested architectures. Degradation follows a concentration-dependent gradient: *catastrophic* on spread models (Qwen 3.31×), *moderate* on flat-high-2bit models (Gemma-2 1.47×, where the solver assigns 106/336=31.5% of heads to 2-bit because distortions are indistinguishable), and *mild* on spike/flat-low-2bit models (Yi 1.08×, Llama 1.07×, Mistral 1.08×).

4.3 Solver Equivalence under Constraint

Table 2 demonstrates that the cardinality constraint is the primary intervention, not the solver. On Qwen-7B, unconstrained greedy (PPL 32.45) and unconstrained MCKP (PPL 36.92) both fail catastrophically. With k=8, both produce *identical* bit allocations ({2:8, 4:100, 8:4}) and identical PPL (11.2312). This equivalence replicates on Qwen-14B (Δ=0.0000).

Table 1: Main results across architectures at 4× compression. **Unconstr.**: MCKP without cardinality constraint. **Constr.**: MCKP with k=8 (or natural k where noted). **Greedy**: RateQuant Algorithm 1 with constraint. All unconstrained solvers underperform uniform 4-bit.

Model	BF16	Uniform 4-bit	Constr. MCKP	Greedy Constr.	Unconstr. MCKP	Shape
Qwen-7B	9.81	15.45	**10.91**	**10.91**†	36.92	Spread
Llama-8B	8.09	8.47	8.45	—	8.63	Spike
Mistral-7B	7.04	7.13	7.16	—	7.57	Flat
Gemma-2-9B	6.03	6.17	6.20	—	8.85	Flat
Yi-1.5-9B	7.02	7.22	7.22	—	7.58*	Spike
Pythia-1.4B	16.49	24.29	24.45	—	31.54	Flat
Qwen-14B	6.87	7.20	7.22	7.22†	7.69	Flat

†Greedy constrained = MCKP constrained (Δ=0.0000). *Estimated from 1.08× degradation ratio.

Table 2: Solver equivalence on Qwen-7B (112 heads, 4× compression). The constraint eliminates solver choice entirely.

Solver	PPL	2-bit heads	Distribution
Greedy unconstrained	32.45	68	{2:68, 4:30, 8:4, 16:10}
MCKP unconstrained	36.92	∼70	—
Greedy + k=8	**11.2312**	8	{2:8, 4:100, 8:4}
MCKP + k=8	**11.2312**	8	{2:8, 4:100, 8:4}
Uniform 4-bit	15.45	0	all 4-bit

4.4 Compounding Characterization

Qwen Compounding Sweep. Table 3 shows PPL as a function of the maximum number of 2-bit heads k on Qwen2.5-7B. The curve is U-shaped: k=8 is the sweet spot (2.4% degradation), while k=64 causes 2.34× and k=112 causes 10.07× degradation.

Cross-Architecture Compounding. Table 4 shows that k=8 is universally safe (<2.5% degradation), but curve shapes differ: Qwen exhibits a U-shape (non-uniform benefit exists at the sweet spot), Llama shows gentle monotonic increase, and Mistral is flat-monotonic (every 2-bit head hurts).

Top-k vs. Random-k. To confirm non-additivity, we compare assigning the k most sensitive heads to 2-bit (top-k) versus k random heads (random-k) on Qwen (Table 5). Both exhibit super-linear compounding, but random-k is worse due to potential selection of correlated heads within the same layer.

4.5 When Does Non-Uniform Allocation Help?

Table 6 shows the sensitivity distribution diagnostic across 9 models. Only Qwen2.5-7B meets both conditions (concentration≥0.50, t_1/t_5 ≤0.70), correctly predicting it as the sole beneficiary of non-uniform allocation. Yi-1.5 (concentration 0.71 but t_1/t_5=0.90) and CodeLlama (0.80, 0.81) illustrate why concentration alone is insufficient: single extreme outliers (spike pattern) do not provide optimization opportunities for MCKP.

4.6 Budget Sweep

Table 7 shows that compression ratio amplifies the benefit of constrained MCKP. At 2×, the budget is sufficient and differences are small. At 4× (the primary operating point), MCKP achieves 29.4%

Table 3: Compounding sweep on Qwen2.5-7B. PPL grows super-linearly beyond k=8.

k	PPL	Ratio to baseline
0	9.81	1.00×
8	10.05	1.02×
16	10.42	1.06×
32	11.60	1.18×
64	22.95	2.34×
112	98.75	10.07×

Table 4: Cross-architecture compounding at k=8. All architectures show $<$2.5% degradation, confirming k=8 as a universally safe threshold.

Model	**k=8 PPL**	**Baseline**	**Degrad.**	**Curve**
Qwen-7B	10.05	9.81	2.4%	U-shaped
Llama-8B	8.23	8.09	1.7%	Monotonic
Mistral-7B	7.06	7.04	0.4%	Monotonic

improvement over uniform. At 8×, uniform 2-bit is catastrophic (PPL 2644) while constrained MCKP achieves PPL 57.6—still high but 45× better.

4.7 Long-Context Evaluation

Table 8 demonstrates that long context amplifies the compounding failure. At 8K tokens, uniform 4-bit degrades to 11.15× baseline (PPL 77.75) while constrained MCKP maintains 1.23× (PPL 8.59)—a 9× gap in degradation rate.

4.8 Downstream Task Evaluation

Table 9 shows that Qwen's PPL improvement translates to consistent downstream gains (+1.25 to +4.00 percentage points, 72.8% average recovery). Table 10 shows that the diagnostic correctly predicts downstream outcomes across all 5 evaluated models. Non-spread architectures show minor systematic degradation (Yi $-$0.77pp, Llama $-$0.99pp, Mistral $-$0.48pp) rather than preservation. ARC-Challenge shows the largest per-task sensitivity to allocation strategy, consistent with its reliance on multi-step reasoning where subtle KV cache errors accumulate (Yi $-$3.24pp, Llama $-$2.65pp).

Multi-seed confidence intervals. Three calibration seeds on Qwen2.5-7B yield PPL 10.99$\pm$0.065 (CV=0.59%), with identical macro allocation (2:8, 4:100, 8:4) across all seeds. Downstream confidence intervals confirm robustness: HellaSwag +4.13pp (std=0.002), PIQA +1.96pp (std=0.002), WinoGrande +1.58pp (std=0.012). ARC-Challenge shows +0.00pp (std=0.004), consistent with this task's higher sensitivity to allocation details.

4.9 Ablation Studies

L0 Exclusion Removal. Removing the hard L0 exclusion rule from the solver produces identical PPL (10.91 in both cases): the profiling signal is strong enough for the solver to automatically assign L0's 4 sensitive heads to 8-bit without domain knowledge.

Profiling Resolution. High-resolution profiling (64×2048) yields PPL 10.91 vs. low-resolution (16×512) PPL 11.23, a 2.9% improvement. High-resolution distinguishes 100 heads vs. 37, but the macro bit distribution is identical (2:8, 4:100, 8:4), confirming the constraint's robustness to profiling quality.

Table 5: Top-k vs. random-k compounding on Qwen2.5-7B. Both exhibit super-linear growth; profiling-based selection (top-k) mitigates but does not eliminate compounding.

k	Top-k	Random-k (mean±std)	Ratio (top)	Ratio (rand)
8	10.04	11.07±0.69	1.56×	1.76×
16	10.42	12.91±0.40	1.56×	2.10×
32	11.60	18.33±2.53	1.78×	2.73×
64	22.95	55.94±9.73	4.89×	8.88×

Table 6: Sensitivity distribution diagnostic. The joint condition (conc.$\geq$0.50 AND $t_1/t_5 \leq$0.70) achieves 9/9 accuracy in predicting non-uniform benefit.

Model	**Conc.**	t_1/t_5	**Shape**	**Predicted**	**Actual**
Qwen-7B	0.61	0.57	Spread	Benefit	+27.3%
Llama-8B	0.57	0.85	Spike	No benefit	+0.2%
Yi-1.5-9B	0.71	0.90	Spike	No benefit	0%
Mistral-7B	0.22	—	Flat	No benefit	−0.4%
Gemma-2-9B	0.23	0.38	Flat	No benefit	−0.3%
Pythia-1.4B	0.24	—	Flat	No benefit	−0.7%
Qwen-14B	0.33	—	Flat	No benefit	−0.3%
Phi-3-med	0.17	0.24	Diffuse	No benefit	—*
CodeLlama	0.80	0.81	Spike	No benefit	—*

*Profiling-only (full pipeline not run due to resource constraints); diagnostic prediction consistent with profiling pattern.

K/V Separation. Treating K and V heads as independent MCKP items yields PPL 11.22 vs. joint K+V PPL 10.91. The joint formulation is superior because budget sharing between K and V naturally discovers that V heads are more robust to quantization.

Oracle Comparison. Brute-force optimal allocation on 8 layer groups shows MCKP is exactly optimal on 5/8 groups, with maximum gap 0.018 PPL (0.18% of baseline) on the remaining 3—position swaps within the same bit budget, confirming constrained MCKP is near-optimal.

Cross-Dataset Calibration. Calibrating on C4 instead of WikiText-2 yields 96.4% allocation agreement (108/112 heads identical) and PPL 11.29 (vs. 10.91 with WikiText-2 calibration), confirming robustness to calibration domain. Cross-domain transfer (WikiText-2→C4 eval) maintains the MCKP advantage: 20.56 vs. 35.42 uniform.

Heuristic Baselines. Table 11 compares constrained MCKP against simpler allocation strategies. MCKP consistently outperforms L0-constrained random allocation and naive positional assignment across all 3 models, confirming profiling-based selection has genuine value on architectures where non-uniform allocation is beneficial.

Mistral k Ablation. On Mistral, PPL increases monotonically with k: k=4→7.165, k=8→7.200, k=12→7.245, k=16→7.311, k=18→7.323 (uniform=7.131). Every additional 2-bit head hurts, confirming flat-distribution architectures should use uniform 4-bit.

4.10 Runtime Analysis

Table 12 shows the latency overhead of mixed-precision KV cache. Constrained MCKP incurs 8.5% decode latency overhead vs. uniform 4-bit due to per-head dynamic dequantization dispatch, with identical memory footprint. The MCKP solver itself requires only 5ms (one-time cost). At long context where the quality gap is largest (8K: 1.23× vs. 11.15× degradation), the latency cost is justified.

Table 7: Budget sweep on Qwen2.5-7B. Compression amplifies non-uniform benefit.

Compression	MCKP PPL	Uniform PPL	Improvement
2×	9.85	10.20	+3.4%
4×	10.91	15.45	+29.4%
8×	57.60	2644	>97%

Table 8: Long-context evaluation on Qwen2.5-7B at 4× compression.

Ctx	BF16	MCKP	Uniform	MCKP ratio	Unif. ratio
2K	9.81	10.91	15.45	1.11×	1.57×
8K	6.97	8.59	77.75	1.23×	11.15×

4.11 KIVI Comparison

KIVI [2] 4-bit with group size 32 and residual length 128 achieves PPL 9.90 on Qwen—better than our constrained MCKP (10.91). However, the two approaches are orthogonal: KIVI optimizes per-channel group quantization granularity (within-bit quality), while we optimize per-head bit-width allocation (cross-bit allocation) and characterize its failure modes. Combining KIVI's quantization with our allocation framework is a natural extension.

5 Conclusion

We show that per-head KV cache quantization on hardware-aligned bit-widths $\{2,4,8,16\}$ has a universal failure mode: 2-bit distortions compound non-additively, causing all unconstrained solvers to underperform uniform 4-bit across 8 architectures. A cardinality constraint ($\leq k$ heads at 2-bit) universally prevents this, making solver choice irrelevant (greedy = MCKP, Δ=0.0000). Under moderate compression, only 1/8 architectures benefits from non-uniform allocation—most models should use uniform 4-bit, checkable via a simple two-metric diagnostic (9/9 accuracy). At long context (8K), the constraint reduces degradation from 11.15× to 1.23× baseline.

Limitations. Our study covers 8 architectures up to 14B parameters; 72B-scale profiling was resource-limited. The diagnostic is validated on n=9 models—directional evidence, not a statistical claim. Runtime overhead (8.5%) may vary with longer sequences. Multi-seed downstream confidence intervals are available only for Qwen.

Future Work. Combining cardinality-constrained allocation with KIVI-style group quantization, extending to token-level eviction policies, and validating on 70B+ models.

References

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[2] Zirui Liu, Jiayi Yuan, Hongye Jin, Shaochen Zhong, Zhaozhuo Xu, Vladimir Braverman, Beidi Chen, and Xia Hu. KIVI: A Tuning-Free Asymmetric 2bit Quantization for KV Cache. *Proceedings of ICML*, 2024.

Table 9: Downstream evaluation on Qwen2.5-7B (0-shot, 4× compression). MCKP recovers 72.8% of the BF16-to-uniform gap on average.

Task	**BF16**	**Uniform**	**MCKP**	**Δ (pp)**
ARC-C	0.529	0.481	0.514	+3.16
HellaSwag	0.783	0.728	0.768	+4.00
PIQA	0.793	0.766	0.779	+1.25
WinoGrande	0.690	0.662	0.682	+1.97

Table 10: Cross-architecture downstream summary (0-shot average, 5 tasks). Diagnostic correctly predicts both PPL and downstream outcomes for all 5 models.

Model	**Shape**	**Δ (pp)**	**Predicted**	**Correct**
Qwen-7B	Spread	+1.92	Benefit	✓
Yi-1.5-9B	Spike	−0.77	No benefit	✓
Llama-8B	Spike	−0.99	No benefit	✓
Mistral-7B	Flat	−0.48	No benefit	✓
Gemma-2-9B	Flat	+0.14	No benefit	✓

Table 11: Allocation baseline comparison across 3 architectures. MCKP outperforms all heuristics; heuristics still beat uniform only on Qwen (spread).

Model	**MCKP**	**L0-rand (med)**	**Naive pos.**	**Uniform**
Qwen-7B	**10.91**	11.52	12.23	15.45
Llama-8B	**8.45**	10.04	9.94	8.47
Mistral-7B	**7.16**	7.59	7.48	7.13

Table 12: Runtime benchmark on Qwen2.5-7B (seq_len=2048).

Method	**Prefill** (ms)	**Decode** (ms/tok)	**Throughput** (tok/s)	**Memory** (MB)
BF16	45.86	33.78	29.6	21
Uniform 4-bit	54.25	42.46	23.6	5.25
Constr. MCKP	57.03	46.09	21.8	5.25

Table 13: KIVI comparison on Qwen2.5-7B (4× compression). Approaches are complementary.

Method	**PPL**	**Approach**
BF16 baseline	9.81	—
KIVI 4-bit (g32, r128)	9.90	Per-channel
Constrained MCKP	10.91	Per-head
Uniform 4-bit (ours)	15.45	Per-head